MILITARY SPECIFICATION

RADAR SETS AN/APS-115B AND AN/APS-122

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipments covered by this specification shall be two airborne search radars which shall provide means for searching for and detecting surface targets. The AN/APS-115B shall be a dual radar set for use where full azimuth coverage must be obtained by a dual (fore and aft) installation. The AN/APS-122 shall be a single radar set for use where the desired azimuth coverage can be obtained with a single antenna.

1.2 Classification - The AN/APS-115B and the AN/APS-122 Radar Sets shall consist of the following items:

ITEM	TYPE DESIGNATION	APPL. PARA.
Antenna (two required for AN/APS-115B)	AS-2146/APS-115	3.5.1
Radar Receiver-Transmitter (two required for AN/APS-115B)	RT-889/APS-115	3.5.2
Antenna Position Programmer	MX-7930B/APS-115	3.5.3
Antenna Control	C-7511A/APS-115	3.5.4
Radar Set Control (two required for AN/APS-115B)	C-7512/APS-115	3.5.5

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.6.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for the previous procurement. However, in lieu of the issue listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

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SPECIFICATIONS

Military

MIL-W-5088C	Wiring, Aircraft, Installation of
MIL-E-5400H	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422E(2)	Testing, Environmental, Aircraft, Electronic Equipment
MIL-I-6181D(3)	Interference Control Requirements, Aircraft Equipment
MIL-C-6781B	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-M-7793C(1)	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for delivery of
MIL-T-18303A	Test Procedures, Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-S-20708C	Synchros, 60 and 400 Cycles, General Specification for
MIL-N-18307D	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment, General Requirements for

Industry

ARINC Report No. 407	Synchro System Manual

STANDARDS

Federal

FED-STD-3(1)	Colors, Aeronautical Lighting

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Military

MIL-STD-454A	Standard General Requirements for Electronic Equipment
MIL-STD-704A	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781A	Test Levels and Accept/Reject Criteria for Reliability of Nonexpendable Electronic Equipment
MIL-STD-785	Requirements for Reliability Program (For Systems and Equipment)
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS-17322D	Meter, Time Totalizing, Miniature Digital, 115 Volt, 400 Cycle
MS-25212C	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions

PUBLICATIONS

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-34	Failure Classification for Reliability Testing, General Requirements for
AR-10	Maintainability of Avionics Equipment and Systems; General Requirements for

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

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3.3 Design and Construction - The equipment shall comply with all the applicable requirements of MIL-E-5400, class 1A equipment, for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 530 pounds for the AN/APS-115B and 301 pounds for the AN/APS-122.

3.3.2 Reliability - The reliability program for the equipment as specified herein shall be in full compliance with MIL-STD-785. On reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipments shall operate either on a continuous or intermittent basis with satisfactory performance for a period of at least 150 hours for the AN/APS-115B and of at least 270 hours for the AN/APS-122 without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipments shall have a total operating life of 5000 hours with scheduled maintenance. Parts requiring replacement by scheduled maintenance, and reliable life of such parts shall be specified by the supplier and reviewed and approved by the procuring activity.

3.3.2.3 Reliability in MTBF (Mean Time Between Failures) - The Equipments, including any built-in-test provisions, shall have 150 hours of mean (operating) time between failures for the AN/APS-115B and 270 hours of mean (operating) time between failures for the AN/APS-122 when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with MIL-M-7793. Meters shall have a range of 9999 hours, and shall be included in the following units:

Unit	Type of Meter
Receiver Transmitter	MS-17322-9
Antenna Position Programmer	MS-17322-9

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3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181 except as noted herein.

3.3.6.1 Short Duration Interference - Interference resulting from manual operation of switch may deviate from the limits of MIL-I-6181 as indicated below:

Maximum Duration	Maximum Recurrence	Deviation Permitted
1 second	Once in 3 minutes	30 db

3.3.7 Provisions for Maintainability - The maintainability program, built-in-test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10.

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3.3.7.1 Modular Design - The module concept shall be used in the internal design of equipments. Circuits shall be designed so that a defective functional loop can be isolated to a module or a group of modules. Module as used herein is defined as any segment of an equipment that can be removed and reinserted without a requirement for soldering (Brazing), welding or other similar techniques.

3.3.7.2 Module Accessibility - Modules shall be arranged so that access to any module does not require removing adjacent modules or parts other than access panels. Access for adjustment and module replacement in avionic equipment shall be from one side (or the front) only. It is permissible to have module trays which are removable from the front of the Receiver-Transmitter unit, and which contain modules accessible from the top. Two side access is also permissible in the receiver section. Waveguide disconnect is permissible from a third side.

3.3.7.3 Fault Isolation Test Points - Test points shall be provided for isolating faults beyond the scope of computer diagnostic test program, BIT (built-in-test), or when other provisions have not been made for fault isolation to the module level. In addition, for those equipments incorporating BIT , test points shall be provided to repair the BIT in the event of failure. The test points shall be readily accessible with the unit in its normal aircraft mounting configuration. If individual test points are not used, the test points shall be grouped in multi-pin receptacles, with caps provided. A diagram of durable construction, showing waveforms, voltages, and other applicable measurements related to each pin, shall be attached to the equipment.

3.3.7.4 Special Support Equipment - Self-test features shall be designed to obviate the need for any special support equipment for preflight check, post flight check, or fault isolation. No special support equipment shall be required except as expressly authorized by the procuring activity. For the purpose of this specification, it is assumed that all peripheral equipment is operating satisfactory and is readily available.

3.3.7.5 Lubrication Requirements - Lubrication requirements shall be held to a minimum.

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3.3.7.6 Lubricant Types - Lubricants other than military approved types, shall not be used except as expressly authorized by the Procuring Activity.

3.3.7.7 Lubrication Provisions - Provisions shall be provided to enable all lubrication to be performed with the unit in its normal aircraft mounting configuration.

3.3.7.8 Maintenance Concept - This avionic system is programmed to a "hands-off" concept of maintenance. Equipments are not scheduled for removal and bench check (except where unavoidable for periodic lubrication of electro-mechanical units) and are maintained by corrective maintenance procedures. Corrective maintenance is accomplished without removing the equipment from its installed position in the aircraft and consists of fault isolation to individual modules, replacement of faulty modules and system checkout. Where feasible, fault isolation will be computer initiated utilizing a diagnostic program; where not feasible, equipments will provide self test features or readily accessible test points to accommodate standard test equipment affording fault isolation to the individual replaceable module. In 95 percent of all functional features (excluding wiring and connectors) it will be possible to isolate faults, repair and checkout the equipment with an MTTR (M_{CT}) of 0.6 hours or less for organizational level corrective maintenance. Organizational level corrective maintenance is defined as the following:

(1) Recognition of a fault

(2) Isolation of a fault to a module or a functional loop

(3) Module replacement

(4) Check out of the equipment

All elapsed time used in MTTR calculations shall be accumulated after the fault has been isolated to the radar set. In addition, the capability for performance of readiness testing will be built into each system, subsystem, and equipment to provide preflight readiness indications and to check performance levels when an out-of-tolerance condition is suspected. Circuit breakers will be used in lieu of fuses whenever required to protect parts and circuits likely to carry an overload due to failures or poor adjustment and should be of types and ratings readily available in the Navy supply system.

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3.3.8 Nomenclature, Nameplates and Identification Marking -Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° C ±5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ±1.0 vac, 400 ±5 Hz, three-phase
	26.0 ±0.5 vac, 400 Hz, single phase (for synchro excitation)
	18 vac (for Annunciator and Advisory lamps) (28 v lamps shall be used)
	5 to 28 vac, 400 Hz, single phase (variable voltage for edge lighting)
	27.5 ±0.5 vdc

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class IA equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve II of Specification MIL-e-5400, except that the console control panel shall meet Curve III.

3.3.10.2 Temperature- The equipment operating extreme temperature range shall be -25 degrees C to +55 degrees C. Equipment shall operate with degraded performance allowed from -54°C to -25°C. The equipment operating extreme temperature range for the antenna assemblies shall be -54°C to +55°C.

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3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 3 minutes under standard conditions and 5 minutes at extreme service (-25°C) conditions. Equipment located outside the pressure hull will operate without degraded performance to -54°C after a 10-minute warm-up.

3.3.12 Input Electrical Power - The equipment shall have the following input power requirements.

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) AC Power (three phase) 115/200 v, category "B," 5800 va

(2) DC Power 28 v, category "B," 7 amperes

3.3.12.2 Lighting Power - Lighting power requirements shall not exceed the following specified amounts:

(1) AC Power (single phase) 5 v to 28 v, category "B" 1 ampere for edge lit panels

(2) DC - 18 v, category "B," 1 ampere for annunciators

3.3.12.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

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3.3.12.4 Standby Power - The equipment shall provide for a standby position whereby rated voltages shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.3.13 Cooling - The design for cooling the equipment shall comply with the requirements of Specification MIL-E-5400. The type of cooling shall be for forced ambient air. Evaluation shall be made in accordance with MIL-T-23103.

3.3.14 Synchro Standards - Synchro circuitry connections shall conform to ARINC Report No. 407. Prior to release of engineering drawings, the supplier shall submit to the procuring activity for approval, a simplified schematic showing all synchro connections which affect external wiring. All drawings shall include complete ARINC notation (e.g., X, Y, Z, H, C, index reference, rotation reference, and for differentials, transformation reference) for all synchro circuitry.

3.3.14.1 Synchros - Design of all synchros shall conform to MIL-S-20708.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and service conditions specified herein.

3.4.1 Operation - The equipment described herein shall provide means for searching for and detecting surface targets with limited simultaneous air target detection capability. A 45 degree azimuth sector scan and a full 360 degree azimuth scan shall be provided at the discretion of the operator. Antenna stabilization signals are accepted from the aircraft navigation computer. To enhance the equipment operation, selectable scan speed of 6 or 12 rpm and transmitter pulse widths of 2.5 or 0.5 microseconds with corresponding pulse repetition rates of 400 or 1600 Hz, respectively, are provided. The pulse repetition rates shall be synchronized with the aircraft power frequency and will vary in accordance with the power frequency variations. Outputs of target range and azimuth information are supplied for distribution to aircraft subsystems.

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3.4.1.1 Frequency Characteristics - The equipment shall be capable of operating over a frequency range of 8.5 to 9.6 gHz and the transmitter shall be manually tunable over this range.

3.4.1.1.1 Frequency Agility Characteristics - The equipment shall be capable of a frequency agility output as well as fixed frequency operation. During agility operation, the rf output frequency of the magnetron varies continuously in a frequency band extending from the mechanically set fixed frequency to a lower frequency. The frequency bandwidth varies from a minimum of 40 MHz to a maximum of 85 MHz over the tunable frequency range. The frequency band is continuously swept at a rate of 75 ±5 Hz.

3.4.1.2 Modulation Characteristics - The transmitter shall be pulse modulated at a line locked pulse repetition rate of 380 to 420 Hz by a pulse width of 2.5 ±0.15 usec, or a pulse repetition rate of 1520 to 1680 Hz by a pulse width of 0.5 ±1.0 usec.

3.4.1.3 Transmitter Power Output - The equipment shall be capable of delivering a minimum of 143 kw peak power measured at the output flange of the receiver-transmitter unit.

3.4.1.4 System Noise Figure - The overall system noise figure shall not exceed 7.5 db when measured at 9.375 gHz and 9.5 db when measured at frequencies of 8.5, 9.0 and 9.6 gHz.

3.4.1.5 Receiver Sensitivity - The minimum detectable signal shall be at least -107 dbm for the receiver when operating in the long pulse mode and -101 dbm when operating in the short pulse mode.

3.4.1.6 Power Capacity - The rf system shall be capable of handling 275 kw peak at sea level atmospheric pressure.

3.4.1.7 System Pressure - Units of the system requiring pressurization shall be pressurized to provide satisfactory operation up to an altitude of 30,000 feet. Units shall be capable of withstanding a maximum pressure of 30 psig, without damage to the system. Each unit in the pressurized system shall be sealed sufficiently as not to lose more than 1 psig in 5 min. when pressurized to 20 psig.

3.5 Detail Requirements

3.5.1 Antenna AS-2146/APS-115 - The antenna unit shall meet the following requirements.

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3.5.1.1 Function - The antenna shall focus and direct the energy furnished by the radar transmitter. The antenna shall be capable of continuous rotation - and when selected, of sector scanning, shall provide an output required for north stabilization of indicator presentation when supplied with proper excitation signal voltages. The antenna shall be tilt-stabilized.

3.5.1.2 Form Factor and Mounting - The form factor for the antenna shall not exceed 45.54 inches high by 45.57 inches wide by 45.57 inches deep. All mounting surfaces shall be in one plane which shall be perpendicular to the vertical axis of the antenna.

3.5.1.3 Weight - The weight of the antenna shall not exceed 84 pounds.

3.5.1.4 Contents - In addition to the radiating system and azimuth drive motor with its associated control circuits, the antenna shall include the azimuth differential synchro, relative bearing synchro, relative bearing resolver synchro, tilt stabilization resolver, tilt synchro and servo unit.

3.5.1.5 Controls - The antenna shall be controlled by the Antenna Control C-7511A/APS-115

3.5.1.6 Pressurization - The RF transmission line systems shall be a part of the pressurized system. Pressurization shall be effective over the reliable operating range of the equipment.

3.5.1.7 Attenuation - The one-way loss for transmitted power through the complete waveguide and rotary joints shall not exceed 0.8 db at any frequency in the 8500-9600 MHz range.

3.5.1.8 Voltage Standing Wave Ratio - The VSWR of the antenna microwave components shall not exceed 1.50 when measured at frequencies of 8500, 8720, 8940, 9160, 9375, and 9600 MHz. Selection of rf components to meet these requirements shall not be necessary.

3.5.1.9 Reflector - The reflector shall be a nominal 45 by 26 inches in size, having the curvature of a paraboloidal surface.

3.5.1.9.1 Polarization - The radiated electric field shall be horizontally polarized when the antenna is mounted normally in an erect attitude.

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3.5.1.9.2 Horizontal Pattern - The 3 db beamwidth measured at any depression or elevation angle within the range specified for the vertical pattern shall be not more than 2.4° when measured at frequencies of 9000, 9375, and 9600 MHz and not more than 2.5 degrees when measured at a frequency of 8500 MHz. Directions of maximum power shall occur at the same azimuth position (within 0.4°) at any depression or elevation angle within the limits specified. Any side lobes shall be down at least 22 db from maximum power when measured at a frequency of 9375 MHz and shall be down at least 18 db from maximum power when measured at frequencies of 8500, 9000, and 9600 MHz.

3.5.1.9.3 Vertical Radiation Pattern - Pencil Beam - The 3 db beamwidth measured at any tilt angle shall not be more than 3.6° when measured at frequencies of 9375 and 9600 MHz and shall not be more than 3.8° when measured at frequencies of 8500 and 9000 MHz. The maximum vector shall not vary by more than 0.4° from the mechanical axis of the antenna. Side lobes in the principal plane shall be at least 22 db below the maximum power at any angle of tilt when measured at a frequency of 9375 MHz and shall be at least 18 db below the maximum power of any angle of tilt when measured at frequencies of 8500, 9000, and 9600 MHz.

3.5.1.9.4 Spoiler Plate - A spoiler plate designed for a cosecant squared pattern shall be provided. The plate shall be normally stowed on the back of the antenna and shall be easily attached to and detached from the parabolic reflector.

3.5.1.9.5 Antenna Gain - The gain of the antenna measured at the first waveguide flange in back of the reflector shall be at least 35 db for pencil beam and 30 db for Csc^2 beam when measured at 9375 MHz and shall be at least 34 db for pencil beam when measured at 8500, 9000, and 9600 MHz.

3.5.1.9.6 Antenna Stabilization - Automatic compensation of the antenna elevation gimbal shall be provided to correct for aircraft pitch and roll maneuvers. This shall be accomplished by supplying the correction signals to the inputs of the elevation servo amplifiers. The azimuth scan reference shall be stabilized to compass north.

3.5.1.9.7 Azimuth Accuracy - The maximum static error of the reflector position in the azimuth plane with respect to the azimuth reference inputs shall not exceed ±0.5 degree.

3.5.1.9.8 Elevation Accuracy - Within the limits specified in 3.5.1.8.9 the maximum static error in the elevation gimbal position with respect to the stabilization inputs shall not exceed ±0.5 degree. The maximum velocity error shall not exceed ±2 degrees when the antenna is scanning at 12 rpm.

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3.5.1.9.9 Elevation Limits - The antenna travel in the elevation gimbal shall be limited to ±30 degrees or greater. Any resultant input signal up to ±60 degrees shall not cause damage to the equipment and the equipment shall operate normally when the resultant signal returns to within ±30 degrees.

3.5.1.9.10 Continuous Rotation Rate - In response to operating signals, the reflectors shall rotate at a constant rate of 6 ±1 rpm or 12 ±2 rpm.

3.5.1.9.11 Sector Width - In response to operating control signals, the antenna beam shall scan the 45° sector width to within ±2°.

3.5.1.9.12 Sector Scan Rate - In response to operating controls, the reflector shall scan the 45° sector at a nominal rate of 36° per second or 72° per second.

3.5.1.9.13 Sector Location - In response to operating control signals, the center of the sector shall be adjustable through 360°.

3.5.1.9.14 Searchlighting Provisions - The antenna reflector shall be capable of being searchlighted; i.e., "stopped" at any position in response to operator control.

3.5.1.9.15 Manual Tilt Control Limits - Beam tilt of the reflector shall cover the least of the range of +10 ±2° elevation to -20 ±2° depression in response to manual operation of the "TILT" control. The tilt servo system shall cause the beam to achieve a predetermined tilt position within ±2° of the setting of the "TILT" control when approached from either direction.

3.5.1.9.16 Index Marks - The rotary base shall contain azimuth index marks at 45° intervals.

3.5.1.10 Azimuth Differential Synchro - A synchro shall be provided, geared such that its rotor rotates through an angle equal to the angular rotation of the reflector in the same direction. Backlash between the synchro rotor and the reflector shall not exceed 0.3 degrees.

3.5.1.11 Relative Bearing Azimuth Synchro - A suitable synchro shall be provided for relative bearing stabilization.

3.5.1.12 Tilt Stabilization Resolver Synchro - A synchro geared so that its rotor rotates in the same direction as the reflector and through an angle equal to the angular rotation of the reflector shall be provided.

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3.5.1.13 IFF Antenna

3.5.1.13.1 Function - The antenna shall radiate and collect IFF information when operating with compatible equipment such as Recognition set, Radar, AN/APX-76.

3.5.1.13.2 Power Handling Capability - The rf components shall be capable of handling a peak power of 4 kw.

3.5.1.13.3 Radiation Pattern Characteristics - With the IFF antenna properly located in respect to the reflector of the antenna unit, it shall exhibit the following pattern characteristics at any frequency from 1010 to 1030 and 1090 to 1110 MHz.

3.5.1.13.3.1 Polarization - The radiated electric field shall be vertically polarized.

3.5.1.13.3.2 Azimuth Plane Beamwidth - The beamwidth at the half power points shall be 23° maximum.

3.5.1.13.3.3 Elevation Plane Beamwidth - The beamwidth of the half power points shall be 45° maximum.

3.5.1.13.4 Antenna Gain - The gain of the antenna shall be at least 12 db over the frequency range of 1010 to 1030 and 1090 to 1110 MHz.

3.5.1.13.5 Voltage Standing Wave Ratio - With the IFF antenna assembled on the antenna unit as an operating system, the VSWR measured on the transmitter side of IFF channel of the azimuth rotary joint shall be as follows:

3.5.1.13.5.1 The VSWR shall not exceed 1.70:1 in the frequency range of 1010 to 1030 MHz and 2.00:1 in the frequency range of 1090 to 1110 MHz.

3.5.1.13.6 IFF and Radar Beam Alignment - In the horizontal or vertical plane, the maximum deviation between the axes of the radar main lobe and the IFF main lobe shall not exceed 3 degrees.

3.5.1.13.7 Interference Between Radar and IFF Equipment - Rejection of signals between the radar and IFF antenna shall be such that optimum use of either system shall not be impaired.

3.5.1.13.8 IFF rf system shall not be pressurized.

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3.5.1.14 Leveling - One or more surfaces, capable of providing areas on which can be placed a spirit level, to establish a plane during installation perpendicular to the axis of azimuth rotation of the reflector, shall be included on the antenna. These areas shall be labeled "REF LEVEL" and shall represent a plane perpendicular to the axis of azimuth rotation within 0.1 degree.

3.5.2 Receiver-Transmitter RT-889/APS-115 - The Receiver-Transmitter shall meet the following requirements:

3.5.2.1 Function - The receiver-transmitter shall generate pulses X-band rf energy and shall detect echoes received from these pulses.

3.5.2.2 Form Factor - The form factor for the receiver-transmitter shall not exceed 16.75 inches high by 28.66 inches wide by 16.76 inches deep.

3.5.2.3 Weight - The weight of the receiver-transmitter shall not exceed 140.0 pounds.

3.5.2.4 Contents - This unit shall contain the circuitry necessary for the generation and detection of the rf pulses. This includes the modulator, magnetron, duplexer, mixer, local oscillator, log IF amplifiers and video section. Associated waveguide plumbing, local oscillator AFC circuits, low-voltage power supplies, and fault isolation circuits are a part of this unit.

3.5.2.5 Frequency - The search transmitting and receiving frequencies of this equipment shall be manually tunable over the range of 8500-9600 MHz. The equipment shall operate on a frequency within this range, selected at random, when delivered.

3.5.2.6 RF Components - The rf components shall consist of the duplexer, receiver mixer, AFC mixer, local oscillator, self-test modulator, and interconnecting waveguide components as required.

3.5.2.6.1 Duplexer - The duplexer shall be of short-slot hybrid design containing one dual TR tube. The voltage standing wave ratio of the duplexer when transmitting shall not exceed 1.25:1. The VSWR of the duplexer in the receiving condition shall not exceed 1.4:1. Both VSWR readings shall be taken at 9375 MHz.

3.5.2.6.2 Load Isolation - A microwave load isolator shall be provided between the magnetron and duplexer, having minimum isolation of 10 db and maximum insertion loss of 0.5 db.

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3.5.2.6.3 Standing Wave Ratio - The voltage standing wave ratio of the receiver-transmitter waveguide components in the transmit direction (TR tube simulated fired) shall not exceed 1.25:1 when measured at frequencies of 8500, 9000, 9375, and 9600 MHz.

3.5.2.7 Receiver Section

3.5.2.7.1 Function - The receiver unit shall amplify the IF signals from the mixer, detect, and provide video signals of the proper polarity and amplitude.

3.5.2.7.2 IF Amplifier Characteristics

(1) IF shall be 60 MHz ±1.0 MHz.

(2) IF bandwidth shall be 2.5 MHz minimum for receiving 0.5 usec pulses and 0.5 MHz minimum for receiving 2.5 usec pulses.

(3) The characteristic response of the receiver shall be logarithmic.

3.5.2.7.3 System Noise Figure - The system noise, when measured with a standard microwave noise source at the antenna connection to the receiver-transmitter shall not exceed 7.5 db when measured at 9375 MHz and shall not exceed 9.5 db when measured at frequencies of 8500, 9000, and 9600 MHz with no selection of components.

3.5.2.7.4 Video Output - The receiver video output shall be of a positive polarity with respect to ground and shall be adjustable. It shall be capable of producing the signal characteristics as specified in 3.5.3.6.1.2.

3.5.2.8 Transmitter Section - The transmitter shall be manually tunable and shall deliver rf in the frequency range of 8500-9600 MHz. The transmitter shall accept a trigger pulse from the synchronizer and furnish an rf output pulse from the magnetron with the following characteristics:

3.5.2.8.1 Long Pulse - The pulse width shall be 2.5 ±0.15 usec when the pulse repetition rate of 400 Hz is selected.

3.5.2.8.2 Short Pulse - The pulse width shall be 0.5 ±0.1 usec when the pulse repetition rate of 1600 Hz is selected.

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3.5.2.9 Automatic Frequency Control - The purpose of the AFC shall be to maintain an IF frequency difference of 60 MHz between the local oscillator and the received signal for the heterodyne reception of signals.

3.5.2.9.1 Manual Tuning Provisions - It shall be possible by means of an external switch to disable the automatic frequency control of the search local oscillator. In conjunction with the AFC manual switch there shall be a knob control which shall tune the local oscillator. There shall be a LOCAL-REMOTE switch and a MANUAL TUNE knob located on receiver-transmitter to accomplish manual tuning at the receiver-transmitter.

3.5.2.10 Fault Isolation - Fault isolation circuits shall be provided to monitor the performance of the receiver-transmitter unit and to allow rapid fault isolation.

3.5.2.10.1 Fault Isolation Switch - A fault isolation switch shall be provided with the following positions:

(1) OFF
(2) W/G PRESS
(3) +20 V
(4) -20 V
(5) THY TRIG
(6) FIL REG
(7) THY/MOD
(8) MAG/MOD
(9) AGILE
(10) AFC XTAL 1
(11) AFC XTAL 2
(12) AFC
(13) STC
(14) RCVR XTAL 1
(15) RCVR XTAL 2
(16) RCVR

3.5.2.10.2 Fault Isolation Meter - A fault isolation meter shall be provided to supply go-no-go indications of the functions selected by the fault isolation switch.

3.5.3 Antenna Position Programmer MX-7930B/APS-115 - The Antenna Position Programmer shall meet the following requirements:

3.5.3.1 Function - This unit provides the electrical signals for scanning the antenna, and the output trigger for the radar set.

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3.5.3.2 Form Factor - The form factor for the Antenna Position Programmer shall not exceed 16.53 inches high by 29.7 inches wide by 8.21 inches deep.

3.5.3.3 Weight - The weight of the antenna position programmer shall not exceed 69.0 pounds.

3.5.3.4 Contents - The Antenna Position Programmer shall contain the azimuth scan programmer, tilt stabilization circuitry, forward and aft servo amplifiers, low voltage power supplies, synchronizer, mode control logic circuitry, and fault isolation circuits.

3.5.3.5 Fault Isolation - Fault isolation circuits shall be provided to monitor the performance of the antenna position programmer unit, azimuth drive assemblies, and tilt drive assemblies, and to provide rapid fault isolation.

3.5.3.5.1 Fault Isolation Switch - A fault isolation switch shall be provided with the following positions:

(1) OFF
(2) +5 VDC
(3) +15 VDC
(4) -15 VDC
(5) SYNCHRONIZER
(6) FWD AZ SERVO AMPL
(7) AFT AZ SERVO AMPL
(8) FWD AZ GEAR ASSY
(9) AFT AZ GEAR ASSY
(10) NO. 1 AZ SCAN PROGRAMMER
(11) NO. 2 AZ SCAN PROGRAMMER
(12) FWD TILT SERVO AMPL
(13) AFT TILT SERVO AMPL
(14) TILT FUNCTION GEN
(15) FWD TILT GEAR ASSY
(16) AFT TILT GEAR ASSY

3.5.3.5.2 Fault Isolation Meter - A fault isolation meter shall be provided to supply go-no-go indications of the functions selected by the fault isolation switch. The system configuration as described in this specification will indicate no-go in positions (7), (9), (10) and (13) of paragraph 3.5.3.5.1.

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3.5.3.6 Electrical Connections - The following connections to external circuits shall be provided.

3.5.3.6.1 Radar Scan Converter Output Signals - The following output signals shall be supplied from the Antenna Position Programmer

3.5.3.6.1.1 Radar Trigger - The radar trigger output measured at the antenna position programmer shall be a positive going pulse of 7.5 ±2.5 volts peak amplitude when terminated in a 95 ohm impedance. The pulse width shall be 1.8 to 2.4 usec when measured at the half amplitude point. The repetition rate of the radar trigger shall be the same as the selected radar pulse repetition rate.

3.5.3.6.1.2 Video Output - The video output shall be of positive polarity. The peak amplitude of maximum video signals measured at the antenna position programmer shall be 2.3 ±0.2 volts when terminated in a 95-ohm impedance. The maximum allowable difference between the forward and aft video signal baseline shall be 0.1 volts. The signal to noise ratio at the video output shall be 4 to 1.

3.5.3.6.1.3 Antenna Bearing - The antenna bearing shall be north-referenced and shall be three-wire synchro signal, 11.8 volts rms maximum. The signal shall be capable of driving one control transformer having a minimum open circuit stator impedance (Zso) of 500 ohms. The antenna synchro signal output will lead the 400 Hz 26 vac reference by a nominal value of 25°.

3.5.3.6.1.4 Antenna Bearing Resolver Output - The antenna bearing shall be heading-referenced and shall be a four-wire resolver output providing sin-cosine information. The primary of the resolver is to be excited by a 1600 Hz square wave generated by the synchronizer.

3.5.3.6.1.5 ANT-AZ ALIGN SWITCH - A toggle switch located on the antenna position programmer shall provide a signal to drive the antenna azimuth gear train to 0.0° when in the "ALIGN" position. "OFF" position of the switch shall allow normal scanning of the antenna. The switch shall have a safety cover such that in order to position the switch to the "ALIGN" position the cover must be manually opened.

3.5.3.6.2 IFF Output Signals - The following output signals shall be supplied to the IFF system from the antenna position programmer unit.

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3.5.3.6.2.1 IFF Trigger - The IFF trigger measured at the antenna position programmer shall be a positive going pulse of 10 to 60 volts when terminated in a 95 ohm impedance. The pulse shall be 0.5 to 5.0 usec wide with a maximum rise time of 0.1 usec and a maximum fall time of 1.0 usec.

3.5.3.6.3 Input Signals - The input signals from the aircraft subsystems to the antenna position programmer unit shall be as follows:

3.5.3.6.3.1 Power Inputs - AC and DC power shall be supplied with characteristics as specified in 3.3.12.

3.5.3.6.3.2 Inputs from Aircraft Subsystems - The following inputs shall be provided to control the radar:

3.5.3.6.3.2.1 High Voltage On - A positive going, 6 microsecond pulse shall be provided to turn the radar high voltage on.

3.5.3.6.3.2.2 High Voltage Off - A positive going, 6 microsecond pulse shall be provided to turn the radar high voltage off.

3.5.3.6.3.2.3 Long Pulse - A positive going 6 microsecond pulse shall be provided to select the 2.5 microsecond pulse width. This signal shall also select the 400 Hz PRF and the 6 RPM antenna scan speed.

3.5.3.6.3.2.4 Short Pulse - A positive going, 6 microsecond pulse shall be provided to select the 0.5 microsecond pulse width. This signal shall also select the 1600 Hz PRF and the 12 RPM antenna scan speed.

3.5.3.6.3.2.5 Signal Characteristics - For the signals specified in 3.5.3.6.3.2.1 through 3.5.3.6.3.2.4 the binary "zero" state level shall be 0 $\pm$0.5 volts and the binary "one" state level shall be 4 +1 volts. The maximum steady state current drawn from a line shall not exceed 1.5 milliamperes. Each signal shall be supplied by a twisted shielded pair cable and shall be terminated at the antenna position programmer unit in a differential amplifier. The lower side of the differential amplifier shall be isolated at least 1500 ohms above ground potential.

3.5.3.6.3.3 Inputs from Aircraft Navigation System - The following inputs shall be supplied from the aircraft navigation system:

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3.5.3.6.3.3.1 Aircraft Heading Input - Two separate aircraft headings with respect to north, shall be supplied to the antenna position programmer unit. Each input shall be a three-wire synchro signal 400 Hz, 11.8 vrms maximum. Each heading signal shall be capable of driving a separate control differential transmitter with each control differential transmitter being loaded by two control transformers. The control differential transmitters shall have a Zso of 95.0 ohms minimum and the control transformers shall have a Zso of 444 ohms minimum. This heading input shall lead the 400 Hz, 26 vac reference by 9 ±1 degrees.

3.5.3.6.3.3.2 Aircraft Pitch Input - The aircraft pitch angle shall be supplied as a three-wire synchro signal, 400 Hz, 11.8 volts rms maximum. The pitch signal shall be capable of driving 5 control transformers in parallel, with each control transformer having a Zso of 500 ohms minimum.

3.5.3.6.3.3.3 Aircraft Roll Input - The aircraft roll angle shall be supplied as a three-wire synchro signal, 400 Hz, 11.8 volts rms maximum. The roll signal shall be capable of driving 5 control transformers in parallel, with each control transformer having a Zso of 500 ohms minimum.

3.5.3.6.3.4 Inputs from IFF System - The following inputs shall be supplied from the IFF system:

3.5.3.6.3.4.1 IFF Return - The IFF return measured at the antenna position programmer shall be a positive-going pulse of 50 ±10 v amplitude when terminated in a 95 ohm impedance. The pulse shall be 2.2 ±1 usec wide with a maximum rise time of 0.1 usec and a maximum fall time of 1.5 usec. The d-c baseline shall be 0 ±1 volt. The delay from the leading edge of the IFF trigger to the leading edge of the IFF return shall be 462 ±1 usec.

3.5.3.6.3.5 Radome Open - A SPDT switch mounted on the airframe shall provide a ground to the antenna position programmer when the aircraft radome is open. When the ground is applied, all power shall be removed from the antenna and radar HV on shall be inhibited for safety of maintenance personnel. The ground should be removed when the radome is securely closed allowing normal system operation. The radome switch shall have a manual over-ride position to allow normal operation of the system while the radome is open.

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3.5.4 Antenna Control Unit

3.5.4.1 Function - This unit shall provide control of the antenna parameters.

3.5.4.2 Form Factor - The form factor for the antenna control unit shall be in conformance with MS25212 with a length of 6.00 inches maximum and a depth of 4.98 inches maximum.

3.5.4.3 Weight - The weight of the antenna control unit shall not exceed 3.2 pounds.

3.5.4.4 Controls - The following operating controls shall be provided on the front panel of this unit:

3.5.4.4.1 "TILT" - A knob control shall permit adjustment of the mechanical tilt angle of the radar antenna with respect to its horizontal reference axis from 20° down to 10° up.

3.5.4.4.2 "SCAN" - A multiple position knob control shall be provided to control the condition of antenna rotation. There shall be three positions, "STOP," "SECTOR," and "FULL." When placed in the "STOP" position, this control shall stop the azimuth rotation of the antenna. In the "SECTOR" position, the antenna shall scan a 45° sector. In the "FULL" position the antenna shall normally scan 360°.

3.5.4.4.3 "ANT HEADING" - A knob control shall permit 360° positioning or "Search Lighting" of the antenna by the operator when the scan switch is in the "STOP" position. It shall permit selection of any azimuth angle about which the antenna will sector scan when the "SCAN" switch is in the "SECTOR" position.

3.5.4.4.4 "STAB OUT" - A two-position toggle switch shall provide for disabling or enabling the automatic tilt stabilization of the antenna reflector. When in the "STAB" position, this control shall engage the automatic tilt stabilization of the antenna reflector. In both positions of the switch, the antenna reflector tilt shall be adjustable by means of the "TILT" control.

3.5.5 Radar Set Control C-7512/115 - The Radar Set Control shall meet the following requirements:

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3.5.5.1 Function - This unit shall provide for control of the radar parameters.

3.5.5.2 Form Factor - The form factor for the radar set control unit shall be in conformance with MS25212 with a length of 8.625 inches maximum, and a depth of 4.98 inches maximum.

3.5.5.3 Weight - The weight of the radar control shall not exceed 4.5 pounds.

3.5.5.4 Controls - The following operating controls shall be provided on the front panel of this unit.

3.5.5.4.1 "WARM UP-STBY-HV ON" - A lighted, push-button switch shall provide normal operation in the "HV ON" position. The "STBY" position shall place the radar in a standby condition and remove the high voltage from the magnetron. Illumination of the "WARM UP" lamp indicates that the radar has not had sufficient warm-up time, and the switch is inoperative until the lamp is extinguished by the system timer.

3.5.5.4.2 "LONG PULSE-SHORT PULSE" - A lighted, push-button switch shall select the 2.5 usec transmitter pulse width and 400 Hz pulse repetition rate and 6 RPM scan rate in the "LONG" position. In the "SHORT" position, the 0.5 usec transmitter pulse width and 1600 Hz pulse repetition rate and 12 RPM scan rate shall be selected. Operation of this switch on either the forward or aft radar set control unit shall control both radars.

3.5.5.4.3 "ANT-DUMMY LOAD" - A lighted push-button switch shall provide for normal operation of the radar antenna in the "ANT" position and shall switch the RF output of the transmitter into the dummy load in the "DUMMY" position.

3.5.5.4.4 "AFC-MAN" - A double-throw switch shall establish automatic frequency control of the local oscillator when in the "AFC" position. When in the "MAN" position, this switch shall permit adjustment of the local oscillator by means of the "MAN TUNE" control.

3.5.5.4.5 "MAN TUNE" - A control shall provide manual frequency control by varying the voltage of the local oscillator provided the "AFC-MAN" switch is in the "MAN" position.

3.5.5.4.6 "RCVR GAIN" - A control shall provide for manual gain of the receiver.

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3.5.5.4.7 "STC" - Two knob controls shall provide for adjustment of the STC (Sensitivity Time Control). The "RANGE" knob shall provide for an exponential bias voltage and the "DEPTH" knob shall provide for the selection of the gain reduction characteristics of the STC voltage.

3.5.5.4.8 "FTC-OFF" - A toggle switch shall provide for normal processing of the radar video when in the "OFF" position. In the "FTC" position, the FTC (Fast-Time Constant) video processing circuitry in the receiver shall be employed to enhance the video presentation.

3.5.5.4.9 "PWR-OFF" - A toggle switch shall turn on the equipment in a standby condition. After the warm-up time has elapsed, the "WARM UP" lamp will be extinguished.

3.5.5.4.10 "AGILE-FIXED FREQ" - A lighted, push-button switch shall select normal radar operation in the "FIXED" position, and actuate the magnetron frequency agility circuitry in the "AGILE" position.

3.5.5.4.11 "VIDEO TEST OFF" - A toggle switch provides an indication on the display when in the "VIDEO TEST" position. This indication will provide the figure of merit of the radar system. The signal to provide this indication will be present at the video output of the receiver.

3.5.5.4.12 Indicators - Two press-to-test failure lamps shall be provided to warn the operator of a radar failure in the R-T or the antenna position programmer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

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(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

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4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the government; it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the initial production tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial

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production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±2 g's

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1*
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

*One out of first ten need not be selected and tested if initial production tests are conducted. Sampling tests are not required when reliability assurance tests are conducted.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the qualification phase and production acceptance (sampling) phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the qualification phase, test level "E" shall be used except low temperature to be -25°C. The accept-reject criteria for test plan IV shall be used.

4.4.3.2 Reliability Production Acceptance Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests," except that all equipments produced shall be tested. Test level "E" of MIL-STD-781 shall be used except low temperature to be -25°C.

4.4.3.2.1 All Equipment Test - Each equipment produced shall be tested for 72 hours. Prior to the 72 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of test plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on test plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc, so that the entire compilation will show other than a reject decision.

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4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc, shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

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(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests. Test time accumulated during the reliability test may be counted toward the life test provided the entire 300 hours are accumulated on a single equipment and equipment selected is in accordance with the table below.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft)
Humidity	Room ambient
AC Voltage	115 ±5 volts (at lowest applicable frequency)
DC Voltage	27.5 ±2.0 volts

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests

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or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for level A shall be determined in accordance with the selection chart in appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment specified herein is intended for use as an airborne search radar, particularly for antisubmarine warfare operations.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing. (See 5.1.)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

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6.6 Associated Equipment - The radar sets must operate properly with the following associated equipment:

(1) AN/APX-76
(2) Indicator, Plan Position, IP-1131/AP
(3) Power Supply, PP-6867?A
(4) Attitude Control System, A/A24G-39
(5) Radar Scan Converter Group, AN/ASA-69
(6) Central Repeater Set AM-4923/A

6.7 This specification is under the cognizance of AIR-5333D4.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-81858(AS) RADAR SETS AN/APS-115B and AN/APS-122

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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